Università degli Studi di Napoli

Federico II

Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione

Statistical Learning and Data Analysis – Module A

SAMPLING DISTRIBUTIONS

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Problem

- An entrepreneur who plans to open a new restaurant would like feedback about **typical per-customer sales** if the restaurant uses a particular dinner menu she has designed.
- She plans to sample **25 friends**, who she believes are comparable to a random sample of people who would attend the restaurant. Each friend will pick the items they would order, keeping in mind their **typical budget for eating out**.
- She will then use the **sample mean** of the 25 bills as an estimate of the **population mean μ**, representing the expected per-customer sales.
- However, she is unsure whether a **sample size of 25** is large enough for the sample mean to provide a good estimate of μ.
- To evaluate this, we conduct a **simulation study** to explore the **sampling distribution** of the sample mean.

Key Assumptions

- The per-customer sales are **right-skewed**, so a **Gamma distribution** is appropriate.
- She expects μ (the mean) to lie between **20 and 24**, with individual sales varying from about 8 to *40*. For the example the case supposed (μ, σ): (20,5) and (24,8)

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Main definitions

- **Population distribution**: This is the distribution from which we select the sample. Some statistical methods assume a particular family of probability distributions, such as normal or binomial. We make inferences about its characteristics, such as the mean μ or probability π.
- **Sample data distribution**: This is the distribution of the n observations in our sample. We describe it by statistics such as the sample mean y and standard deviation s. As n increases, the sample data distribution more closely resembles the population distribution.
- **Sampling distribution**: This is the probability distribution for the possible values of a sample statistic, such as y. A sampling distribution determines the probability that the statistic falls within a certain distance of the population parameter that it estimates

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Chebyshev's inequality

- For any r.v. X with $E(X) = \mu$ e $Var(X) = \sigma^2 < +\infty$, for any given $\varepsilon > 0$ it holds:

$P(|X - \mu| < \varepsilon\sigma) \geq 1 - \frac{1}{\varepsilon^2}$

- Assume $\varepsilon\sigma = k$, Chebyshev's inequality can be also expressed as for any $k > 0$:

$$P(|X - \mu| < k) \geq 1 - \frac{\sigma^2}{k^2}$$

- For any random variable X of an ***unknown probability model*** but ***known mean and variance***, it is always possible to find the minimum level of probability such that the variable takes values belonging to the interval $[\mu - \varepsilon\sigma, \mu + \varepsilon\sigma]$.

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Convergence Laws

- **Convergence in distribution:**

$$X_1, \dots, X_n, \dots \xrightarrow{d} X \text{ if } F_n(X) \xrightarrow{d} F(X)$$

- **Convergence in probability:**

$$X_n \xrightarrow{p} X \text{ if } \forall \varepsilon > 0 \text{ it holds: } \lim_{n\to\infty} P(|X_n - X| < \varepsilon) = 1$$

- **Strong Law of large numbers:** Performing the same experiment a large number of times, the average of the results should be close to the expected value and will tend to be closer as more trials are performed. The sample average converges *almost surely* to the expected value.
- **Weak Law of large numbers:** The sample average converges *in probability* to the expected value.

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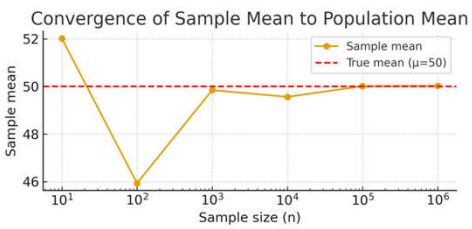
Convergence in Probability



Concept:
- Involves random variables rather than fixed values.
- The sample mean Y converges in probability to the population mean μ:
  $Y \rightarrow p\ \mu$

Chebyshev's Inequality and the Law of Large Numbers:

As $n \rightarrow \infty$, this probability $\rightarrow$ 0 (Law of Large Numbers)

Python simulation:

```
import numpy as np
np.mean(np.random.uniform(0,100,10))      # ≈ 23.6
np.mean(np.random.uniform(0,100,1000))    # ≈ 51.9
np.mean(np.random.uniform(0,100,10000000)) # ≈ 50.0
```

As n increases, Y approaches μ = 50.
Rate of convergence: $SE = \sigma/\sqrt{n} \rightarrow |Y - \mu|$ rarely $> 3\sigma/\sqrt{n}$.

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De Moivre Laplace and Central Limit Theorems

- **De Moivre Laplace Theorem:** Let $X_1, X_2, \dots, X_n$ be n Bernoulli random variables, independent and identically distributed (i.i.d.) with mean π and variance $\pi(1-\pi)$. As n tends to infinite, the sum of n Bernoulli variables tends to a normal distribution with mean π and variance $n\pi(1-\pi)$, as also

$$\frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\pi, \frac{\pi(1-\pi)}{n}\right)$$

- **Central Limit Theorem:** Let $X_1, X_2, \dots, X_n$ be n random variables, independent and identically distributed (i.i.d.) with mean μ and variance σ^2. As n tends to infinite, the sum of all variables tends to a normal distribution with mean μ and variance $n\sigma^2$, as also

$$\frac{1}{n}\sum_{i=1}^{n} X_i \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

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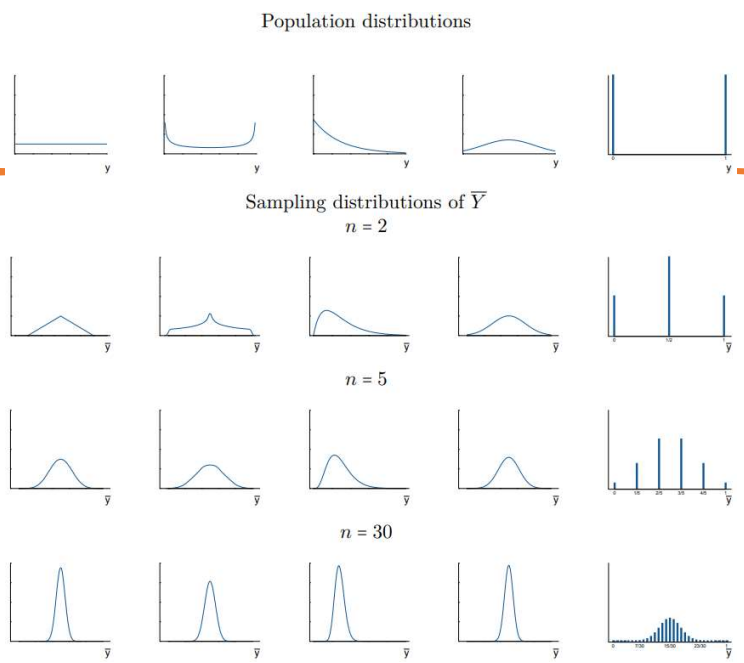

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Sum of Independent Random Variables

In many probability models, the sum of independent random variables preserves the family of the distribution.

This property is crucial for deriving compound distributions and for applications in statistical inference.

- **Normal Distribution**: If Y_i are independent normal random variables, with $Y_i \sim N(\mu_i, \sigma_i^2)$:

$$\Sigma_i Y_i \sim N(\Sigma_i \mu_i, \Sigma_i \sigma_i^2)$$

The sum is also normal, with means and variances adding.

- **Binomial Distribution:** If Y_i are independent binomial random variables, with $Y_i \sim Binomial(n_i, \pi)$:

$$\Sigma_i Y_i \sim Binomial(\Sigma_i n_i, \pi)$$

The sum is also binomial, with number of trials adding likewise, if the success probability π is the same.

- **Poisson Distribution:** If Y_i are independent Poisson random variables, with $Y_i \sim Poisson(\mu_i)$:

$$\Sigma_i Y_i \sim Poisson(\Sigma_i \mu_i)$$

The sum is also Poisson, with means also adding.

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